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Tuesday

COI - AT3 - Descriptive test

CSA1717 - Artificial Intelligence

Unit - 1

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- 1) A smart home automation system uses artificial intelligence to automatically control lighting, temperature and security based on user behaviour and environmental conditions. The system receives real-time inputs from sensors such as motion detectors, temperature sensors, cameras and clocks. The agent analyze this information, make decisions and perform actions.

Simple Reflex agent:

A simple reflex agents acts only on the current percept using predefined if-then rules. It does not remember previous events or predict future situations.

Eg: If motion is detected → turn on lights

Advantages:

- ★ simple & fast
- ★ Low computational cost.

★ suitable for straight-forward tasks.

Model-Based reflex agent:

A model-based reflex agent maintains an internal model of the environment (memory). It combines current sensor data with previous information for making decisions.

Examples:

If windows are open, adjust air conditioning accordingly.

Advantages:-

- ★ Better decision-making
- ★ Avoids unnecessary actions.

Disadvantages:

- ★ more complex
- ★ requires memory & additional processing.

Goal-Based agent:

A Goal-based agent chooses actions that help achieve a specific goal.

Goals in Smarthome:

- ★ maintain room temperature.
- ★ Reduce electricity consumption.
- ★ Ensure home security.
- ★ Improve convenience.

Eg: Turn on outdoor lights when someone approaches at night.

Advantages:

- ★ make intelligent decisions.
- ★ Flexible in changing environments.

Limitations:

- ★ Planning requires more computation.

Utility based Agent:

A Utility-Based agent selects the action that provides the highest overall benefit by considering various factors.

Utility Factors: User comfort, energy efficient, security, cost of electricity, user preferences.

Eg: Lock doors & activate cameras when house is

Advantages:

- ★ Balances multiple objectives.
- ★ optimal decisions.
- ★ learns & adapts over time.

Limitations:

- ★ More expensive computationally.

Recommended Approach:

A hybrid agent combining Model-Based, Goal-based and utility based approach is most effective solution for a smart home automation system. Because it is more adaptable than all the agents. It provides best performance, remember past events, planning, decision making according to environmental conditions.

- 2) A robot placed in an unknown maze must explore different paths until it reaches the exits. Since it has no prior map, it uses uninformed search strategies which don't require heuristic knowledge.

Uniform Cost Search:

UCS expands the node with the lowest path cost first.

- ★ The robot keeps tracks of the cost of every path explored.
- ★ If some paths are longer or more expensive than others, UCS always chooses the cheapest path available.
- ★ It guarantees the optimal path when all steps costs are non-negative.

Advantages:

- ★ It is optimal.

Disadvantages:

- ★ High memory usage.

Iterative Deepening search: (IDS)

It repeatedly performs Depth-limited search with increasing depth limits.

First search depth 0, then depth 1 and so on.

Advantages:

- ★ Low space complexity.

Disadvantages:

- ★ Not optimal when path costs are different.
- ★ more time complexity.

Intelligent agents:

Simple reflex agent:

- ★ acts only on the current perception.
- ★ Eg: Turn left when wall is ahead.
- ★ May get stuck in loops.

Model-based agent:

- ★ maintains an internal map of visited locations.
- ★ Avoids revisiting same cells.
- ★ Better for unknown mazes.

Goal-Based agent:

- ★ It has a goal and works towards the goal.
- ★ Eg: Goal = reach the exit.
- ★ Uses search algorithms such as UCS or IDS to plan actions.

Utility Based Agent:

- ★ considers multiple factors such as path length, energy consumption & safety.
- ★ chooses the actions with the highest utility.

Most effective combination:

The Goal-based + Model-based agent with Uniform Cost Search will be the best choice.

- ★ Goal based: focuses on reaching exit.
- ★ model-based: remembers explored paths & avoids loops
- ★ UCS: Finds the least cost path.

A model-based & Goal-based agent with UCS is the most effective as it combines memory, planning, and optimal path selection, better decision making.